

A disproof of Conjecture 00000002307

Fitting height versus derived length

1 The conjecture

Conjecture 00000002307 reads:

Definition: The Fitting height and derived length are two invariants of solvable groups.

Conjecture: Fitting height $\leq 2 \cdot (\text{derived length}) - 1$ (a linear bound); the bound is controlled by layerwise projections of supersolvable towers; tightness is attained by extremal examples with total derived chain 2.

It is the conjunction of three clauses. Writing $h(G)$ for the Fitting height and $\text{dl}(G)$ for the derived length:

- (i) (*linear bound*) $h(G) \leq 2 \text{dl}(G) - 1$ for every finite solvable group G ;
- (ii) the bound is “controlled by layerwise projections of supersolvable towers”;
- (iii) (*tightness*) the bound is attained by an example of derived length 2, i.e. some solvable group G has $\text{dl}(G) = 2$ and $h(G) = 2 \text{dl}(G) - 1 = 3$.

Clause (ii) is not a precise mathematical statement, and we do not need it: the conjunction $(i) \wedge (ii) \wedge (iii)$ is false as soon as (i) or (iii) is false. We show that **both (i) and (iii) are false**. Clause (iii) fails for a structural reason (Theorem 4: $h(G) \leq \text{dl}(G)$ for every solvable group), so the refutation does not depend on how the trivial group is treated.

2 Definitions

We use the standard definitions, as stated for instance on the Wikipedia pages “Fitting length” (https://en.wikipedia.org/wiki/Fitting_length) and “Solvable group” (https://en.wikipedia.org/wiki/Solvable_group):

Definition 1. A *Fitting chain* (Fitting series, nilpotent series) of length n of a group G is a subnormal series

$$1 = H_0 \trianglelefteq H_1 \trianglelefteq \cdots \trianglelefteq H_n = G$$

whose successive quotients H_{i+1}/H_i are nilpotent. The *Fitting length* (Fitting height, nilpotent length) $h(G)$ is the smallest length of a Fitting chain.

“A Fitting chain ... for a group is a subnormal series with nilpotent quotients ... The Fitting length or nilpotent length of a group is defined to be the smallest possible length of a Fitting chain” (Wikipedia, *Fitting length*).

Definition 2. The derived series is $G^{(0)} = G$, $G^{(k+1)} = [G^{(k)}, G^{(k)}]$. The group is solvable if $G^{(n)} = 1$ for some n , and the *derived length* $\text{dl}(G)$ is the least such n .

“The least n such that $G^{(n)} = 1$ is called the derived length of the solvable group G ” (Wikipedia, *Solvable group*). With these definitions the trivial group has $h(1) = \text{dl}(1) = 0$: the chain $1 = H_0 = G$ has length 0, and $G^{(0)} = G = 1$.

3 Fitting height is at most derived length

Lemma 3. Let $B \leq A$ be subgroups of a group G with $B \trianglelefteq A$ and $[A, A] \leq B$. Then A/B is abelian, hence nilpotent.

Proof. For $x, y \in A$ we have $(xy)^{-1}(yx) = y^{-1}x^{-1}yx = [y^{-1}, x^{-1}] \in [A, A] \leq B$, so $xyB = yxB$. Abelian groups are nilpotent (of class ≤ 1). $\square$

Theorem 4. For every solvable group G (finite or not), $h(G) \leq \text{dl}(G)$.

Proof. Let $d = \text{dl}(G)$, so $G^{(d)} = 1$. Put $H_i = G^{(d-i)}$ for $0 \leq i \leq d$. Then $H_0 = G^{(d)} = 1$ and $H_d = G^{(0)} = G$. Each $G^{(k)}$ is normal in G (it is characteristic), and $G^{(k+1)} = [G^{(k)}, G^{(k)}] \leq G^{(k)}$; hence $H_i \leq H_{i+1}$ and $H_i \trianglelefteq H_{i+1}$. Writing $A = H_{i+1} = G^{(k)}$ and $B = H_i = G^{(k+1)}$ with $k = d-i-1$, we have $[A, A] = B$, so H_{i+1}/H_i is nilpotent by Lemma 3. Thus $(H_i)_{0 \leq i \leq d}$ is a Fitting chain of length d , and $h(G) \leq d$. $\square$

Corollary 5. Let G be a solvable group.

- (a) If $G \neq 1$ then $\text{dl}(G) \geq 1$, hence $h(G) \leq \text{dl}(G) \leq 2 \text{dl}(G) - 1$: clause (i) holds for every nontrivial solvable group.
- (b) If $h(G) = 2 \text{dl}(G) - 1$ then $\text{dl}(G) \leq 1$. In particular no solvable group of derived length 2 has $h(G) = 3$.

Proof. (a) If $\text{dl}(G) = 0$ then $G = G^{(0)} = 1$. (b) $2 \text{dl}(G) - 1 = h(G) \leq \text{dl}(G)$ gives $\text{dl}(G) \leq 1$. If $\text{dl}(G) = 2$ then $h(G) \leq 2 < 3$. $\square$

4 The refutation

Theorem 6. (1) Clause (i) is false: the trivial group is finite and solvable with $h = \text{dl} = 0$, and $0 > 2 \cdot 0 - 1$.

- (2) Clause (iii) is false: by Corollary 5(b), no solvable group (finite or infinite) of derived length 2 attains the bound, which would require Fitting height 3.

Consequently Conjecture 00000002307 is false.

Remark 7. Part (2) is the substantive refutation. Even if clause (i) is read only for nontrivial groups, where it is true (Corollary 5(a)), the claimed extremal examples “with total derived chain 2” cannot exist: equality $h = 2 \text{dl} - 1$ occurs exactly for the nontrivial abelian groups ($h = \text{dl} = 1$). For example S_3 has $\text{dl} = 2$ and $h = 2$, and S_4 has $\text{dl} = 3$ and $h = 3$; in both cases $h \leq \text{dl}$, and the bound $2 \text{dl} - 1$ (equal to 3 and 5) is not attained.

5 Lean 4 formalization

The directory `lean/` contains a Lean 4 project (Lean v4.33.1, Mathlib v4.33.1). The single source file `Conjecture2307/Basic.lean` contains:

- `FittingChain G n`: subgroups $H\ 0 = \perp \leq \dots \leq H\ n = \top$, each normal in the next ($(H\ i).subgroupOf\ (H\ (i+1))$ is normal), with `Group.IsNilpotent` of the quotient group $H_{i+1}/H_i\ (H\ (i+1)\ /\ (H\ i).subgroupOf\ (H\ (i+1)))$;
- `fittingLength G` := `sInf {n | Nonempty (FittingChain G n)}` and `derivedLength G` := `sInf {n | derivedSeries G n =  $\perp$ }` (Mathlib's `derivedSeries`);
- `LinearBound` (clause (i), over all finite solvable $G : \text{Type}$, inequality in $\mathbb{Z}$) and `TightAtDerivedLengthTwo` (clause (iii));
- `fittingLength_le_derivedLength` (Theorem 4), `linearBound_of_nontrivial` and `derivedLength_le_one_of_tight` (Corollary 5), `not_linearBound`, `not_tightAtDerivedLengthTwo` (Theorem 6), and
- the main theorem

`conjecture_00000002307_false :  $\neg$ (LinearBound  $\wedge$  TightAtDerivedLengthTwo).`

There is no sorry, no `native_decide` and no added axiom; `#print axioms conjecture_00000002307_false` reports only `propext`, `Classical.choice` and `Quot.sound`. To check:

```
cd lean
lake exe cache get
lake build
```